

ORION-RSE: Recursive Scientific Governance as a Falsifiable Decision Contract

ORION-P14

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Abstract

Research agents can generate hypotheses, retrieve literature, design experiments and draft interpretations faster than they can reliably determine what those outputs scientifically warrant. We present **ORION-RSE**, a fail-closed recursive scientific-governance contract that separates research generation from claim promotion. ORION-RSE atomizes claims, subtracts nearest-donor ownership before novelty promotion, freezes protected discriminators before outcomes, retains null/negative/subsumed history, distinguishes standalone from interaction-only evidence, records **CANNOT_CHECK**, separates evaluator identity from scientific authority, stops recursion when no material discriminator remains and reopens prior decisions only after material evidence or regime change. P14A is retained as a negative mixed benchmark because the decisive discriminator occupied only 1.8375% of realized cases and aggregate gates could not pass. P14B creates a balanced semantic discriminator but is diagnostic because the full policy reused the same adjudication function as gold and because hostile review found a protocol conformance mismatch. P14C is the specification-separated successor: 28 explicit gold cases are frozen separately, gold/rationale/case identity are stripped before policy calls, the full implementation is independent, and six component ablations are tested. Full ORION-RSE is 28/28 correct with zero false promotions and full valid-discovery recall; **MULTI_REVIEW** is 24/28 with 14.29% false promotion, all ablations are worse, and two evaluations produce identical canonical SHA-256 74032348...f01a63. The evidence supports a precise controlled claim: **against the registered governance specification, the complete contract conforms strictly better than partial review contracts without suppressing valid promotion**. External scientific validity remains open.

1 Introduction and donor boundary

Scientific research is not equivalent to producing a plausible hypothesis or a positive result. A claim must be interpreted relative to prior work, the question frozen before observation, negative/null evidence, competing explanations, resource constraints and evaluator authority. Research agents amplify both opportunity and failure: they can produce more candidate studies, but also more false novelty, widened claims, post-hoc endpoints, forgotten negatives, donor duplication and confident closure when evidence is non-identifying.

Autonomous and semi-autonomous systems already search literature, generate hypotheses, run code, design experiments and revise research artifacts. Goal-evolving scientific agents make recursive discovery an explicit objective. Reflection, self-critique, reviewer agents and multi-agent debate own iterative review; preregistration/checklists own prospective endpoint discipline; truth-maintenance systems own dependency-aware history; provenance owns evidence lineage; authorization systems own separation of actor permission from operation.

ORION-RSE claims none of those primitives. Its question is:

Can an explicit scientific-governance contract make better claim-promotion decisions than strong partial review contracts under matched information and decision resources, while preserving valid discoveries?

The live residual is the **scientific promotion contract plus its evaluation**: claim atomization, donor subtraction, protected freeze, negative/subsumed history, interaction/non-identifiability dispositions, authority separation, recursion stops and material-change reopening composed into one fail-closed lifecycle and evaluated on false-promotion, disposition and useful-discovery endpoints.

2 ORION-RSE contract, lifecycle and evaluation

A research atom carries a bounded claim/question, parent identity, nearest donors, protected discriminator, protocol identity, evidence/resource receipts, evaluator identity, authority owner, disposition and stop/reopen conditions. Atomization is an investigation unit, not automatic publication fragmentation.

The contract distinguishes `SUPPORTED_RESIDUAL`, `SUBSUMED`, `INTERACTION_ONLY`, `REDUNDANT_EQUIVALENT`, `OVERREACH_HARMFUL`, `NON_IDENTIFIABLE`, `CANNOT_CHECK`, negative/null outcomes and `RETAIN_NEGATIVE` when new material does not justify reopening. Positive observation is therefore not identical to positive scientific promotion.

Before novelty promotion, nearest prior mechanisms are recorded and the residual is classified as adopted, adapted, composed, deferred or subsumed. The discriminator and terminal rules are frozen before protected outcomes; a material post-outcome change requires a new protocol identity. Negative results can motivate successors but cannot be rewritten.

The lifecycle is:

question -> atoms -> donor subtraction -> protected discriminator -> protocol freeze -> execution/receipts -> disposition -> recurse/stop -> later reopen if materially warranted.

`CANNOT_CHECK` is a valid scientific disposition when evidence or authority is insufficient.

2.1 Evaluation contract

All policies receive identical case facts and identical fixed decision-check budgets. Registered policies are `RAW_POSITIVE`, `REFLECTION_CHECKLIST`, `DONOR_AWARE_REVIEW`, `MULTI_REVIEW`, `ORION_RSE_FULL` and component ablations. Primary safety endpoint is false scientific promotion; productivity endpoint is useful-discovery recall among gold `SUPPORTED_RESIDUAL` cases; full disposition accuracy and component-specific correctness are secondary. This prevents an always-abstain system from winning.

3 P14A — preserved negative mixed benchmark

P14A used 20 held-out families $\times$ 400 cases with independently varied fact rates. Full ORION-RSE produced false promotion 0.000000, useful-discovery recall 1.000000, disposition accuracy 1.000000 and history/reopen accuracy 1.000000.

The strongest comparator, `MULTI_REVIEW`, produced false promotion 0.018375, useful-discovery recall 1.000000, disposition accuracy 0.981625 and history/reopen accuracy 0.505051.

The frozen protocol required strongest-baseline false promotion ≥ 0.05 and accuracy separation ≥ 0.08 . Both gates failed. Terminal: `P14A_CONTROLLED_GOVERNANCE_SUPERIORITY_GATE_NOT_MET`. This terminal is permanent.

3.1 Root cause

After validity, freeze, identifiability, donor and interaction checks, the only difference between `MULTI_REVIEW` and full ORION-RSE is live negative history without material new evidence. In the realized random mixture, that effective discriminator occupied only **1.8375%** of cases, so the maximum possible aggregate accuracy gap was also 1.8375 points. The result identifies a benchmark-design problem: natural/random mixtures may underweight the decision boundary being tested. P14A is not returned; it motivates a balanced successor.

4 P14B — balanced semantic discriminator, diagnostic only

P14B uses a fresh seed and 12 held-out families with seven equal strata: clean support, legitimate material reopening, retained negative history, donor subsumption, interaction-only evidence, `CANNOT_CHECK` and negative evidence. There are 6,720 protected cases total with the same seven-check receipt for every arm.

Full ORION-RSE yields zero false promotion, 1.0000 disposition accuracy and 1.0000 useful-discovery recall. `MULTI_REVIEW` yields 0.142857 false promotion, 0.857143 disposition accuracy and 1.0000 useful-discovery recall. Both systems correctly allow genuinely material reopening; only full ORION-RSE retains negative history when no material new evidence exists. Every registered component ablation is worse.

P14B is not authoritative primary evidence. First, its full policy originally invoked the same decision function used as protected gold, making the implementation comparison circular. Second, hostile PR review found the realized generator did not implement the protocol's promised nuisance reminting across all strata. The benchmark is retained as semantic-development history and carries no primary claim authority.

5 P14C — specification-separated primary benchmark

P14C was frozen after the P14B circularity issue was identified and does not alter P14A or P14B.

- `P14C_ADJUDICATION_CASES_V1.json` contains **28 explicit frozen cases**, four variants for each of seven semantic strata.
- `run_p14c_specification_separated_governance_v1.py` implements every policy independently from the case table.
- Before a policy call, the harness strips `gold_disposition`, `rationale`, `case_id` and `stratum`; the policy receives only factual booleans.
- Precedence variants test donor-over-interaction/history, interaction-over-history, validity failure and negative evidence.
- Six ablations separately remove evidence-integrity, freeze, identifiability, donor, interaction and negative-history checks.

The explicit specification is internal to the programme, not a human external-adjudication dataset. P14C therefore evaluates **conformance to a separately frozen specification**, not truth about open-ended science.

Terminal: P14C_SPECIFICATION_SEPARATED_GOVERNANCE_CONFORMANCE_SUPPORTED.

policy	disposition accuracy	false promotion	useful-discovery recall
ORIONRSEFULL	1.0000	0.0000	1.0000
MULTI_REVIEW	0.857143	0.142857	1.0000
DONOR_AWARE_REVIEW	0.714286	0.285714	1.0000
REFLECTION_CHECKLIST	0.571429	0.428571	1.0000
RAW_POSITIVE	0.428571	0.535714	1.0000

Full ORION-RSE correctly handles all retained-negative and supported-reopen cases. All six ablations lower disposition accuracy. Gold is absent from every policy input. Two independent evaluations produce identical canonical SHA-256 74032348de7e6508b6c1827aabc1bf9d354d30b9c6f81c8259fdb3535f

The strongest current claim is: **against an explicit adjudication specification frozen separately from policy implementation, the full ORION-RSE contract conforms to every registered governance case and strictly outperforms the registered partial-governance implementations without reducing valid promotion.** This is stronger than a self-call conformance test and weaker than external scientific validity.

6 Required external validation and resource accounting

A broad claim that ORION-RSE improves real research requires prospectively assembled realistic packets across multiple domains such as ML experiments, formal/software verification and computational science. Candidate workflows should receive identical literature/source packets, tools and model budgets. Independent blinded adjudicators must determine the maximum admissible claim, donor overlap, negative-evidence handling, CANNOT_CHECK and valid discoveries.

Comparators should include a strong research agent, reflection/self-critique, researcher-reviewer multi-agent workflow, literature-RAG plus experiment planning, preregistration/checklist control and ORION component ablations.

Longitudinal evaluation should create positive, null, negative and subsumed history in round 1, then change evidence, donor set or regime in round 2. The system must avoid both forgetting old negatives and over-transferring old negatives after genuine material change. Useful-discovery noninferiority remains co-primary so safety cannot be purchased by blanket abstention.

Every workflow should report model/checkpoint, evidence access, web/literature access, tool access, generated tokens, experiment/search budget, reviewer/evaluator calls, number of research/reflection passes and end-to-end latency. ORION-RSE cannot hide extra compute as “governance.” Final scientific authority remains external to candidate workflows.

7 Relation to current research-agent work, limitations and conclusion

Goal-evolving and autonomous scientific systems establish open-ended generation as donor-owned. Research on scientific-exploration behavior shows that evaluation should consider what agents choose to explore, not only whether they can produce plausible papers. ORION-RSE studies a complementary layer: **scientific admissibility governance**. Its distinctive mechanism is active negative/subsumed history plus material-reopen semantics on top of validity, donor and interaction checks.

P14A–C are controlled symbolic decision benchmarks, not natural-language open-ended science. P14C removes direct gold-function reuse but its specification is still internally authored; external

validity remains open. Balanced strata test discrimination, not real-world prevalence; P14A shows why prevalence and discriminability must be reported separately. Real workflows have heterogeneous costs. Human or independent-model adjudicators may disagree on novelty and admissibility. Longitudinal value of negative-history retention is not yet shown on realistic research. Poorly calibrated reopen/materiality criteria could suppress productive risk-taking, so useful-discovery noninferiority is mandatory. No claim of frontier autonomous-research superiority is authorized.

ORION-RSE treats scientific governance as a decision system that can itself fail. P14A fails because its decisive discriminator is too rare; P14B is semantically informative but non-authoritative; P14C removes implementation circularity with a separately frozen case specification and independent policy implementation. The complete contract conforms to all registered cases, makes zero false promotions and preserves every valid promotion, while the strongest partial review contract fails the retained-negative boundary.

The next claim frontier is external rather than synthetic: **blinded realistic packets, independent adjudication and longitudinal research history under matched resources.**

8 References

- Du, Y. et al. *Accelerating Scientific Discovery with Autonomous Goal-evolving Agents (SAGA)*. arXiv:2512.21782.
- Tang, Y. & Yang, Y. *AI Research Agents Narrow Scientific Exploration*. arXiv:2605.27905, 2026.
- Related reflection, multi-agent review/debate, preregistration, truth-maintenance, provenance and authorization literatures are donor-owned foundations and are discussed in the integrated manuscript.